

Arrow's impossibility theorem



In any society, some people might want extra resources to go to schooling, and others may want it spent on roads. When making collective decisions, societies must grapple with the problem of such conflicting desires. Arrow's impossibility theorem shows that logically there is no procedure for making such decisions that simultaneously satisfies a small number of reasonable criteria for fairness. One of these properties is that there is no 'dictator': no single person who determines society's decision. Another is that if *every* individual preferred schools to roads, then society as a whole should prefer schools to roads. The theorem states that these, and other apparently sensible criteria for fairness, are inconsistent.

Consider a voting system that met the second condition: as a result there would be a dictator – a single vote would always determine the outcome. The theorem is powerful and

disturbing: even the most transparent voting systems are likely to be bedevilled by inconsistency and paradox.